Rocket Reusability Analysis

Supporting SpaceY's Competitiveness Through Data-Driven Insights

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Github link: https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/tree/main

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Executive Summary

This report applies the CRISP-DM methodology to analyse historical rocket launch and landing data with the objective of supporting SpaceY's ambition to compete with SpaceX through increased first-stage reusability and lower launch costs. Reusability plays a critical role in reducing cost per launch and improving long-term operational efficiency.

Exploratory analysis, geospatial visualization, and predictive modelling were used to identify factors associated with successful rocket reuse. While predictive models achieved consistent performance—approximately 83% accuracy for three out of four models—the limited number of historical launches and improving success rates over time constrain confidence in selecting a single best model and suggest a high likelihood of future model drift.

Despite these limitations, the analysis provides robust and actionable insights. Recovery-related design features such as rocket legs and grid fins emerged as the strongest drivers of reuse success. The results demonstrate that the primary value of the analysis lies in guiding engineering and operational focus, rather than attempting to precisely predict individual launch outcomes.

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1. Introduction

Space launch costs remain a major barrier to scalable access to space. SpaceX has demonstrated that first-stage rocket reuse can significantly reduce these costs, creating a strong competitive advantage. SpaceY seeks to pursue a similar strategy by improving its ability to safely and consistently recover and reuse rocket stages.

The purpose of this report is to use data-driven analysis to better understand which factors influence reuse success and how these insights can support SpaceY's strategic and operational decisions. Rather than focusing solely on model performance, the analysis is framed around enabling better engineering investment and reuse-oriented decision-making.

2. Methodology (CRISP-DM)

This analysis follows the CRISP-DM framework:

Business Understanding

Define the business objective: improve first-stage reuse to lower cost per launch and enhance competitiveness.

Data Understanding

Historical launch and landing data were examined to understand mission characteristics, launch sites, landing outcomes, and time trends.

Data Preparation

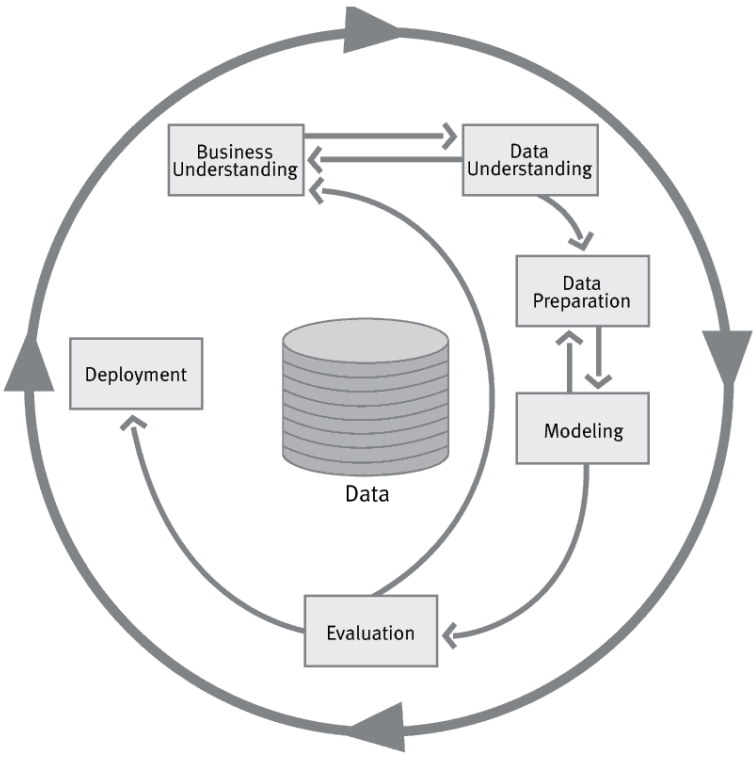
Data cleaning, transformation, and feature engineering were applied, including encoding categorical variables and defining reuse success as the target outcome.

Modelling

Four classification models were trained and evaluated to explore reuse prediction, with a focus on interpretability and consistency rather than optimization.

Evaluation

Models were compared using accuracy and qualitative agreement of insights. Outputs were assessed based on business relevance rather than predictive metrics alone.



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3. Exploratory Analysis & Visualization

Exploratory data analysis revealed that reuse success does not depend on a single variable, but on a combination of factors including mission type, payload characteristics, launch site, and recovery conditions.

Interactive Folium maps were used to visualize launch locations and landing zones, revealing operational patterns and geographic clustering relevant to recovery strategies. Additionally, time-based analysis showed a clear improvement in reuse success over successive launches, indicating learning and process improvement effects.

Figure 1: Number of successful launches per location

4. Modelling Results

Four classification models were trained to predict first-stage reuse success. The logistic regression, K-nearest neighbours (KNN), and support vector machine (SVM) models each achieved approximately 83% accuracy, demonstrating consistent predictive performance across different modelling approaches. The decision tree model exhibited more variable performance, which is expected given its sensitivity to data splits and its tendency to capture noise when trained on relatively small datasets.

However, the overall number of available launches remains limited, reducing statistical confidence in identifying one model as definitively superior. Furthermore, the observed increase in reuse success over time suggests that model assumptions may become outdated as operational practices improve, creating a risk of model drift.

Importantly, interpretation of the logistic regression model highlighted rocket legs and grid fins as variables with the strongest absolute coefficient values. This finding consistently aligns with domain expectations and reinforces the importance of recovery-enabling hardware.

```
yhat=logreg_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```

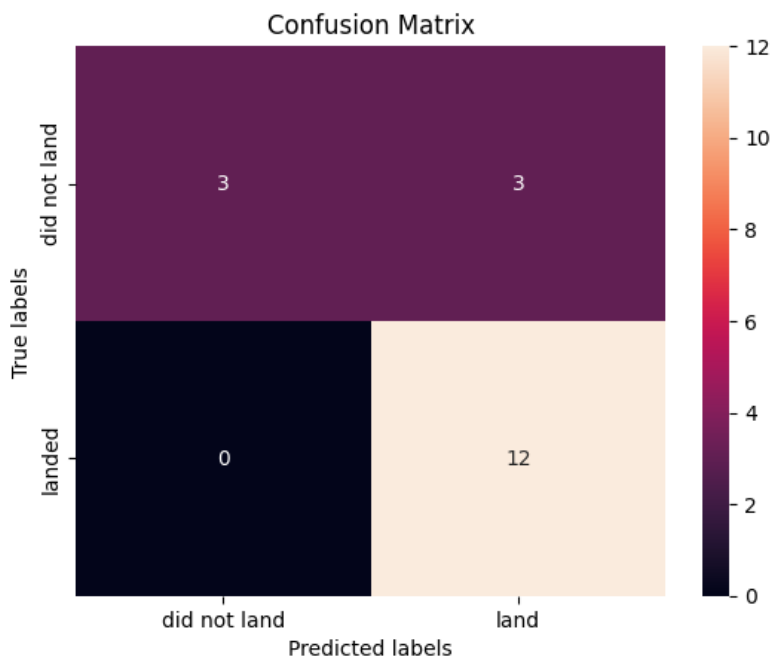

Figure 2: Confusion matrix of the logistic regression model

5. Discussion

The results indicate that predictive modelling can support reuse strategy decisions but should not be treated as a standalone solution. The consistency of findings across models increases confidence in the identified drivers of reuse success, even when overall predictive certainty remains limited.

The analysis confirms that improvements in recovery systems—rather than marginal tuning of mission parameters—are central to reuse success. Additionally, increasing reuse rates over time highlight the need for continuous model reassessment as SpaceY's capabilities evolve.

6. Recommendations

Based on the analysis, the following recommendations are proposed:

- Prioritize investment in recovery hardware, particularly rocket legs and grid fins, as key enablers of reuse
- Use predictive models as decision-support tools, not as definitive predictors of launch outcomes
- Regularly retrain and reassess models to mitigate model drift caused by operational learning
- Combine data insights with engineering expertise and operational feedback loops to sustain reuse improvements
- Model a Recurrent Neural Network (RNN) with time sequence modeling
- Expand feature data set to improve models' accuracy

7. Conclusion

This analysis demonstrates that data science can meaningfully support SpaceY's reuse strategy when anchored in a clear business objective. While limitations in data volume and temporal dynamics constrain predictive certainty, consistent insights across multiple models provide valuable guidance on where to focus engineering and operational efforts.

By emphasizing recovery systems and continuously adapting analytical models, SpaceY can strengthen its ability to reduce launch costs and enhance long-term competitiveness.

Acknowledgements

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Appendices

- All codes used and data tables can be found in the GitHub repository:

https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/tree/main

- Microsoft Copilot (GPT-5 based) was used as a support tool for coding assistance and text refinement (Final decisions, conclusions, and recommendations are author-owned)